

Manish Pandey

Curriculum Vitae

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RESEARCH INTERESTS

Probability theory; stochastic processes; random graphs; large deviations; associative memory models.

ACADEMIC POSITIONS

Aarhus University <i>Postdoctoral Researcher</i>	Denmark <i>July-2025 - Present</i>
Eindhoven University of Technology (TU/e) <i>Ph.D. in Mathematics; Supervisor: Prof. Remco van der Hofstad</i>	Netherlands <i>Aug 2020 – Mar 2025</i>

EDUCATION

Indian Statistical Institute, Delhi and Kolkata <i>M.Stat. in Statistics (Probability Specialization); Overall: 89.5%</i>	India <i>Jul 2018 – Jun 2020</i>
Indian Statistical Institute, Bangalore <i>B.Math. (Hons.); Overall: 89.4%</i>	India <i>Jul 2015 – May 2018</i>
St. Theresa Senior Secondary School, Haldwani <i>AISSCE (Class XII); Overall: 93%; Science aggregate: 96.34%</i>	India <i>Apr 2014 – Mar 2015</i>

PUBLICATIONS

1. A. Chakrabarty, **M. Pandey**, S. Chakrabarty, *Length of stationary Gaussian excursions*, *Proceedings of the American Mathematical Society* **151**(3): 1339–1348, 2023.
2. R. van der Hofstad, **M. Pandey**, *Connectivity of random graphs after centrality-based vertex removal*, *Journal of Applied Probability* **61**: 967–998, 2024.
3. R. van der Hofstad, **M. Pandey**, *Are giants in random digraphs “almost” local?*, *Electronic Communications in Probability* **30**: 1–13, 2025.
4. O. Nagy, **M. Pandey**, G. Exarchakos, M. Bentum, R. van der Hofstad, *Communication protocol for a satellite-swarm interferometer for low-frequency radio astronomy*, *Acta Astronautica* **246**: 821–831, 2026.
5. M. A. Ramos-Docampo, C. Duan, N. Wang, **M. Pandey**, P. de Dios Andres, C. Hirsch, B. M. Städler, *Modelling active particle motion from fluorescence correlation spectroscopy data*, *Analytical Chemistry*, 2026, **98**(31): 22781–22791.

PREPRINTS AND SUBMITTED MANUSCRIPTS

1. S. Dhara, **M. Pandey**, Leonard J. Schulman, *Cascade surviving cores via local limit*, *arXiv:2609.16242*, submitted 2026.
2. G. Exarchakos, R. van der Hofstad, O. Nagy, **M. Pandey**, *Bringing order to network centrality measures*, *arXiv:2601.16236*, submitted, 2026.

THESIS AND EXPOSITORY WRITING

Ph.D. thesis: *Centrality measures and connectivity properties in large networks*, Eindhoven University of Technology, 2025.

M. Pandey, *Centrality Measures: Who Is the Most Important in a Network?*, *The Network Pages*, 2023.

HONOURS AND AWARDS

- ISI Teachers' Committee Award for outstanding academic performance in the M.Stat. programme (2018–2020).
- ISI Teachers' Committee Award for outstanding academic performance in the B.Math. (Hons.) programme (5 of 6 semesters, 2015–2018).
- Institute stipend/fellowship throughout the M.Stat. (2018–2020) and B.Math. (Hons.) (2015–2018) programmes.
- Nominee for the P. C. Mahalanobis Gold Medal (Top 4 in the M.Stat. cohort, Indian Statistical Institute).
- International Rank 41 and Zonal Rank 1 (Uttar Pradesh and Uttarakhand), SOF International Mathematics Olympiad, 2015.
- All India Rank 9, M.Stat. Entrance Examination, Indian Statistical Institute.
- Top 1% in NSEP (2014–2015); Merit Certificate, IAPT.

RESEARCH VISITS

University of Augsburg

Visiting Researcher

March 2026

Department of Mathematics

Purdue University

Visiting Research Scholar

Summer 2024

School of Industrial Engineering

TEACHING AND SUPERVISION EXPERIENCE

Aarhus University

Lecturer: Random Networks

Spring 2026

Department of Mathematics

Aarhus University

Lecturer: Monte Carlo Simulation

Fall 2026

Department of Mathematics

Aarhus University

Co-supervisor of Ph.D. student

Aug 2025 – Present

Department of Mathematics

Eindhoven University of Technology
Teaching Assistant: Calculus, Probability, and Statistics

Aug 2020 – Feb 2024
Department of Mathematics

INVITED TALKS AND PRESENTATIONS

University of Augsburg <i>Invited Speaker</i>	Mar 2026 <i>Department of Mathematics</i>
Concluding NETWORKS Conference <i>Invited Speaker</i>	Oct 2025 <i>The Netherlands</i>
ICTS-NETWORKS Workshop “Challenges in Networks” <i>Poster Presentation</i>	29 Jan–02 Feb 2024 <i>ICTS Bengaluru, India</i>
LMS Probability Research School <i>Poster Presentation</i>	26–30 Jun 2023 <i>University of Liverpool</i>
NETWORKS Training Week <i>Speaker</i>	May 2022; Oct 2022 <i>Asperen, the Netherlands</i>
Institut Henri Poincaré, Paris <i>Invited Speaker</i>	Jul 2021 <i>Bézout–Eurandom Conference / Franco–Dutch Meeting</i>
Master Classes in Probability <i>Invited Participant</i>	Jan 2020 <i>University of Strasbourg</i>

INTERNSHIPS

State Street Global Advisors <i>Active Quantitative Equity Intern</i>	Summer 2019 <i>Bangalore, India</i>
NovelSynth Soft Solutions (Petrabytes) <i>Research Intern, Machine Learning</i>	Winter 2017 <i>Bangalore, India</i>

ONGOING PROJECTS

Temporal motif counts in dynamic random graphs; Centrality comparison curves; Hopfield model in geometric random graphs; Locality properties of centrality measures; Large-deviation analysis of sparse many-body systems in motion; Topological data analysis in zebra-fish enteric nervous system.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, R, HTML.
Tools and Software: $\text{\LaTeX}$, MATLAB, Microsoft Office.

LANGUAGES

English; Hindi.